

FACE RECOGNITION ATTENDANCE SYSTEM USING TRADITIONAL MACHINE LEARNING
A COMPUTER VISION PROJECT REPORT

BACHELOR OF SCIENCE IN DATA SCIENCE AND ANALYTICS

Advanced Machine Learning

Abstract

This project presents the design and implementation of a real-time Face Recognition Attendance System using traditional computer vision and machine learning techniques. The system employs Haar Cascade for face detection, Histogram of Oriented Gradients (HOG) for feature extraction, and a Support Vector Machine (SVM) for face recognition to automate student attendance efficiently and accurately.

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1. Introduction

Face recognition has become one of the most widely adopted applications of computer vision due to its ability to automatically identify individuals from digital images and video streams. The technology has found practical applications in education, healthcare, banking, security, surveillance, access control, and workforce management because it provides a reliable and contactless means of identity verification (Szeliski, 2022). Unlike traditional attendance methods that rely on manual registers, identification cards, or biometric fingerprint scanners, face recognition enables attendance to be recorded without physical interaction while significantly reducing opportunities for impersonation and administrative errors.

Attendance management plays a critical role in educational institutions because it provides evidence of student participation, supports academic administration, and contributes to institutional decision-making. However, conventional attendance recording methods remain inefficient, particularly in large classes where lecturers spend considerable time verifying student presence. Manual systems are also susceptible to proxy attendance, missing records, transcription errors, and delayed reporting, all of which affect the integrity of attendance data (Jain, Ross, & Nandakumar, 2011).

Recent developments in computer vision have made automated attendance systems increasingly practical. A typical face recognition system consists of several sequential stages including image acquisition, preprocessing, face detection, feature extraction, classification, and identity verification (Bradski & Kaehler, 2008). The quality of recognition depends not only on the classification algorithm but also on the effectiveness of the preprocessing and feature extraction techniques used to represent facial characteristics. Consequently, designing a robust attendance system requires careful integration of each stage within the computer vision pipeline rather than relying on classification alone.

Traditional machine learning techniques continue to provide effective solutions for many face recognition problems despite the rapid advancement of deep learning. Histogram of Oriented Gradients (HOG) has been widely recognized as an effective handcrafted feature descriptor because it captures local edge orientations that remain relatively stable under moderate illumination changes and facial expression variations (Dalal & Triggs, 2005). Similarly, Support Vector Machines (SVMs) have demonstrated strong classification performance in high-dimensional feature spaces due to their ability to maximize the margin between different classes while maintaining good generalization performance (Cortes & Vapnik, 1995). These characteristics make the combination of HOG and SVM particularly suitable for small and medium-sized datasets where computational efficiency and interpretability are important considerations.

Face detection forms the foundation of any face recognition system because only correctly detected facial regions should be forwarded for feature extraction and classification. Among the traditional detection techniques available, the Haar Cascade classifier introduced by Viola and Jones (2001) remains one of the most influential methods for real-time object detection. Its use of Haar-like features, integral images, and cascade classifiers enables rapid detection while maintaining acceptable accuracy for frontal face recognition applications. Although more sophisticated deep learning detectors have emerged in recent years, Haar Cascade continues to provide an appropriate solution for lightweight applications that require fast execution and limited computational resources.

The development of this project followed an incremental design approach. Initial experiments were conducted using the ORL Face Database to establish a complete understanding of the traditional face recognition pipeline, including preprocessing, feature extraction, classification, and evaluation. While the ORL dataset provided a useful benchmark for validating the methodology, it did not adequately represent the practical conditions encountered in classroom attendance systems, where variations in illumination, camera positioning, and live image acquisition are unavoidable. To address this limitation, the project evolved into a fully customized real-time attendance system that acquires facial images directly from a webcam, constructs a personalized dataset, retrains the recognition model using the collected images, and performs live attendance recording through facial recognition.

The final system was implemented entirely in Python using OpenCV for image acquisition, preprocessing, and Haar Cascade face detection; Scikit-learn for feature scaling, Support Vector Machine classification, and model evaluation; NumPy for numerical computation; Pandas for attendance record management; Pillow for graphical image handling; Matplotlib for visualization; and Tkinter for developing the desktop graphical user interface. Each library was selected because of its maturity, extensive documentation, computational efficiency, and compatibility with traditional computer vision workflows. Together, these technologies enabled the development of a complete end-to-end

attendance management system while remaining fully aligned with established computer vision principles and avoiding the use of deep learning models.

The completed system provides an integrated platform through which users can register students, capture facial images, train the recognition model, perform real-time attendance using a webcam, maintain attendance records, and monitor system performance through an intuitive graphical interface. Using a custom dataset comprising **450 facial images collected from 9 registered students**, the system demonstrates that traditional computer vision techniques remain capable of delivering accurate, efficient, and practical face recognition solutions for education attendance management.

2. Problem Statement

Accurate attendance management remains a persistent challenge in many educational institutions due to the continued reliance on manual recording methods. Traditional attendance registers require lecturers to allocate valuable instructional time to calling students' names or circulating attendance sheets. Besides reducing teaching time, these approaches are susceptible to human errors such as incorrect recording, omission of student names, and delayed data entry. More importantly, manual attendance systems are vulnerable to impersonation, where one student signs or responds on behalf of another, thereby compromising the credibility of attendance records (Jain et al., 2011). Although biometric technologies such as fingerprint and iris recognition have been introduced to address some of these shortcomings, they require physical contact with sensing devices, increasing hardware costs and reducing convenience when serving large numbers of users. Contact-based systems also experience challenges related to sensor wear, hygiene concerns, and increased maintenance requirements (Li & Jain, 2015).

Face recognition offers a contactless alternative that enables attendance to be recorded automatically as individuals appear before a camera. However, many existing face recognition systems rely on deep learning architectures that require extensive computational resources, specialized hardware, and large annotated datasets for effective training (Goodfellow, Bengio, & Courville, 2016). Such requirements make them unsuitable for learning environments where computational resources are limited or where traditional computer vision techniques are specifically required.

Furthermore, many publicly available face datasets, including the ORL Face Database, are captured under relatively controlled laboratory conditions. While these datasets are useful for algorithm development and benchmarking, they do not adequately represent the practical conditions encountered during real-world attendance recording, where variations in lighting, camera positioning, facial expressions, and background conditions are common. Consequently, models trained solely on public datasets may experience reduced performance when deployed in live environments.

This project addresses these challenges by developing a real-time face recognition attendance system using traditional computer vision techniques. The proposed solution employs Haar Cascade classifiers for reliable face detection, Histogram of Oriented Gradients (HOG) for robust handcrafted feature extraction, and a Support Vector Machine (SVM) classifier for facial recognition. Rather than relying solely on a public dataset, the system incorporates a custom image acquisition module that captures facial images directly from a webcam, allowing the recognition model to be trained using images collected under the actual operating conditions of the intended application. This approach improves the practical relevance of the system while maintaining computational efficiency and compliance with the project's restriction against the use of deep learning techniques.

The resulting system provides an automated, contactless, and efficient attendance management solution capable of reducing administrative workload, minimizing attendance fraud, and improving the accuracy and reliability of attendance records within educational environments.

3. Objectives

3.1 General Objective

The general objective of this project was to design and implement a real-time face recognition attendance system using traditional computer vision and machine learning techniques to automate attendance recording accurately, efficiently, and without physical contact.

3.2 Specific Objectives

The study sought to achieve the following specific objectives:

1. To acquire facial images of registered students using a webcam and develop a custom image dataset suitable for training a face recognition model.
2. To preprocess the captured facial images through grayscale conversion, image resizing, Gaussian noise reduction, and histogram equalization in order to improve image quality and ensure consistent feature extraction.
3. To detect human faces in both captured images and live video streams using the Haar Cascade object detection algorithm.
4. To extract discriminative facial features using the Histogram of Oriented Gradients (HOG) descriptor, thereby transforming facial images into numerical feature vectors suitable for machine learning.
5. To train and optimize a Support Vector Machine (SVM) classifier capable of accurately recognizing registered students based on the extracted HOG features.
6. To evaluate the performance of the developed recognition model using standard classification metrics, including accuracy, precision, recall, F1-score, false positive rate, false negative rate, specificity, confidence scores, and processing time.
7. To develop a desktop application that integrates student registration, dataset creation, model training, real-time attendance recognition, attendance record management, and system configuration into a single platform.
8. To demonstrate that traditional computer vision techniques can provide an effective and computationally efficient solution for automated attendance management without relying on deep learning algorithms

4.0 Literature Review

4.1 Introduction

Face recognition has emerged as one of the most significant applications of computer vision due to its ability to automatically identify individuals from digital images and video streams. Over the past two decades, advances in image processing, feature extraction, and machine learning have transformed face recognition from a purely academic research topic into a practical technology deployed across education, healthcare, banking, border control, surveillance, and access management systems (Szeliski, 2022). Within educational institutions, face recognition has attracted considerable attention because it provides a contactless, efficient, and secure approach to attendance management while reducing the administrative burden associated with manual attendance recording.

This chapter reviews the theoretical concepts, algorithms, and previous studies relevant to the development of the proposed face recognition attendance system. The discussion begins with computer vision fundamentals before examining face recognition, image preprocessing, Haar Cascade face detection, Histogram of Oriented Gradients (HOG), Support Vector Machines (SVM), and previous research on automated attendance systems.

4.2 Computer Vision

Computer vision is a branch of artificial intelligence that enables computers to acquire, process, analyze, and interpret visual information from images and videos in a manner similar to human vision (Szeliski, 2022). The objective is not merely to capture images but to extract meaningful information that can support automated decision-making.

A conventional computer vision system follows several sequential stages including image acquisition, preprocessing, object detection, feature extraction, classification, and evaluation. Each stage contributes to improving the quality of the final prediction. Poor performance in one stage often affects subsequent stages, making the overall pipeline

dependent on the effectiveness of each component (Gonzalez & Woods, 2018). Unlike deep learning approaches that automatically learn image representations from large datasets, traditional computer vision relies on handcrafted features that explicitly describe important characteristics such as edges, textures, shapes, and gradients. Although deep learning has become increasingly popular, handcrafted feature extraction methods remain computationally efficient and suitable for applications with relatively small datasets or limited computing resources. The present study adopts the traditional computer vision workflow because it satisfies the project requirements while providing a transparent and interpretable recognition pipeline.

4.3 Face Recognition

Face recognition is a biometric technology that identifies or verifies an individual's identity by analyzing distinctive facial characteristics. The process generally consists of face detection, facial feature extraction, and identity classification (Jain et al., 2011). Face recognition has several advantages over other biometric technologies. Unlike fingerprint or iris recognition, facial recognition does not require physical contact with specialized sensors. Individuals can be recognized at a distance using ordinary cameras, making the technology particularly suitable for classroom attendance management and access control systems. However, reliable face recognition remains challenging because facial appearance changes with illumination, facial expressions, ageing, pose variation, occlusion, and camera quality. Consequently, effective preprocessing and feature extraction are essential for improving recognition accuracy before classification is performed (Zhao et al., 2003). The current project addresses these challenges by integrating image enhancement techniques with robust handcrafted feature extraction before classification.

4.4 Image Processing

Image preprocessing improves image quality before feature extraction and classification. The objective is to reduce unwanted variations while preserving important facial characteristics required for recognition (Gonzalez & Woods, 2018). Several preprocessing operations were implemented in this study. The captured facial images were first converted into grayscale. Color information contributes little to structural facial representation while increasing computational complexity. Grayscale conversion therefore simplifies subsequent processing and reduces computational requirements.

Each detected face was then resized to **128 × 128 pixels**. Standardizing image dimensions ensures that every sample produces feature vectors of identical length, which is necessary for successful machine learning classification. Gaussian Blur was subsequently applied to suppress random image noise generated by the webcam sensor. Noise reduction improves feature stability by smoothing minor intensity fluctuations without significantly altering facial structures. Histogram Equalization was finally performed to improve image contrast by redistributing pixel intensities across the available grayscale range. This process enhances important facial details and reduces the effect of uneven lighting conditions, thereby improving the robustness of feature extraction.

4.5 Image Detection

Face detection represents the first stage of any face recognition system because feature extraction can only be performed after facial regions have been correctly located. Viola and Jones (2001) introduced the Haar Cascade object detection algorithm, which remains one of the most influential traditional face detection techniques. The algorithm employs Haar-like features computed rapidly through integral images and organizes weak classifiers into a cascade structure that rejects non-face regions while preserving computational efficiency. Compared with exhaustive pixel-by-pixel scanning, the cascade classifier significantly reduces processing time because simple classifiers eliminate obvious non-face regions before more complex classifiers are evaluated.

The present study employed OpenCV's pre-trained **haarcascade_frontalface_default.xml** classifier. This detector was selected because it provides reliable frontal face detection, requires no additional training, integrates seamlessly with OpenCV, and supports real-time webcam applications. Although more sophisticated detectors exist, Haar Cascade remains appropriate for lightweight systems that prioritize speed, simplicity, and compliance with traditional computer vision methodologies.

4.6 Histogram of Oriented Gradient (HOG)

Feature extraction transforms facial images into numerical representations that can be processed by machine learning algorithms. Dalal and Triggs (2005) introduced the Histogram of Oriented Gradients (HOG) descriptor to represent local object appearance through gradient orientation distributions. Rather than analyzing raw pixel intensities, HOG focuses on edge directions and local texture information, making it relatively robust to moderate illumination changes and facial expression variations.

In this project, each preprocessed facial image was divided into small cells where gradient orientations were computed and aggregated into histograms. The resulting histograms were normalized across neighboring blocks before being concatenated into a single feature vector containing **10,800 numerical descriptors** for every facial image. HOG was selected because it provides discriminative facial representations while maintaining relatively low computational requirements. Its compatibility with Support Vector Machines has also been extensively validated in previous face recognition studies.

4.7 Support Vector Machine

Support Vector Machines are supervised machine learning algorithms designed for classification and regression problems. Cortes and Vapnik (1995) proposed SVMs as margin-based classifiers that identify the optimal separating hyperplane between different classes.

Unlike many classification algorithms, SVM seeks to maximize the margin separating neighboring classes, thereby improving generalization on previously unseen samples. Kernel functions further enable SVMs to classify data that cannot be separated linearly. The Radial Basis Function (RBF) kernel was employed in this study because preliminary experimentation demonstrated superior performance compared with alternative configurations. Hyperparameter optimization identified $C = 0.1$ and $\text{Gamma} = 0.01$ as the optimal configuration for the final recognition model. SVM was selected because it performs exceptionally well in high-dimensional feature spaces generated by HOG descriptors while remaining computationally efficient for medium-sized datasets such as the custom dataset developed in this project.

4.8 Previous Studies on Face Recognition Attendance Systems

Numerous studies have investigated automated attendance systems using facial recognition. Arsenovic et al. (2017) developed a classroom attendance monitoring system that automatically recognized students using traditional machine learning techniques, demonstrating substantial reductions in attendance processing time. Similarly, Turk and Pentland (1991) introduced the Eigenfaces approach, one of the earliest appearance-based face recognition techniques. Although modern handcrafted descriptors such as HOG have largely surpassed Eigenfaces in robustness, the study established the importance of feature representation in facial recognition. More recently, many attendance systems have adopted deep learning models because of their high recognition accuracy. However, these systems generally require extensive computational resources, powerful graphical processing units, and large annotated datasets for effective training (Goodfellow et al., 2016). Such requirements increase development complexity and may not be practical in resource-constrained educational environments. Unlike these approaches, the present study demonstrates that carefully integrated traditional computer vision techniques comprising Haar Cascade face detection, HOG feature extraction, and Support Vector Machine classification remain capable of achieving high recognition accuracy while maintaining real-time performance using standard computing hardware.

4.9 Literature Review Summary

The reviewed literature demonstrates that successful face recognition systems depend on the effective integration of image preprocessing, reliable face detection, discriminative feature extraction, and accurate classification. Previous research confirms the effectiveness of Haar Cascade for real-time face detection, HOG for handcrafted facial representation, and Support Vector Machines for high-dimensional classification tasks.

However, many existing studies either rely on controlled public datasets or employ computationally intensive deep learning models. This project addresses these limitations by developing a real-time attendance system trained on a custom dataset acquired under realistic operating conditions while employing only traditional computer vision techniques. Consequently, the developed system contributes a practical, computationally efficient, and academically compliant solution for automated attendance management.

5. Methodology

5.1 Introduction

This chapter describes the methodology adopted in developing the Face Recognition and Attendance System. A traditional computer vision pipeline was implemented, beginning with image acquisition and progressing through preprocessing, face detection, feature extraction, classification, performance evaluation, and deployment within a desktop application. The selected methodology ensured compliance with the project requirements by avoiding deep learning techniques while demonstrating the complete workflow of a machine learning-based computer vision system.

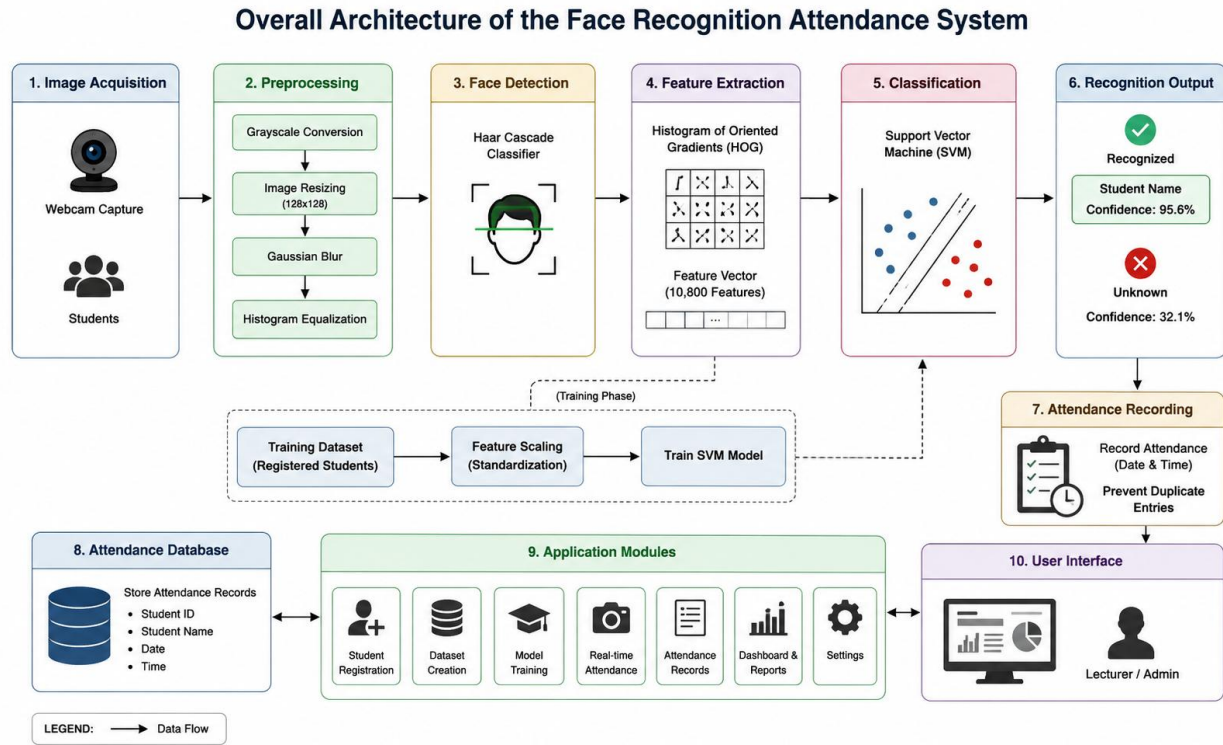
The development process was iterative. Initial experiments were conducted using the ORL Face Database to understand and validate the face recognition pipeline under controlled conditions. Although the ORL dataset provided an excellent foundation for algorithm development, it did not adequately represent the practical challenges associated with real classroom attendance. Consequently, the project was redesigned to use a custom dataset acquired through a webcam, allowing the final model to learn facial characteristics under realistic operating conditions.

5.2 System Development Approach

The project followed a sequential computer vision workflow comprising the following stages:

1. Image acquisition.
2. Image preprocessing.
3. Face detection.
4. Feature extraction.
5. Feature normalization.
6. Machine learning classification.
7. Attendance recording.
8. Performance evaluation.
9. Graphical User Interface development.

Figure 1: System Development Approach



Each stage was implemented independently before being integrated into the complete attendance system. This modular approach simplified testing, debugging, and future maintenance because improvements could be introduced into individual modules without affecting the entire application.

5.3 Software and Development Environment

The system was developed entirely in Python because of its extensive support for computer vision, machine learning, and graphical user interface development. Python also provides a rich ecosystem of scientific libraries that simplify image processing while maintaining high computational efficiency.

Table 5.1 summarizes the software tools used during development.

Table 5.1 Development Tools

Tool	Purpose	Justification
Python	Programming language	Simple syntax, extensive scientific ecosystem and excellent library support.
OpenCV	Image processing and face detection	Provides efficient implementations of image processing algorithms and Haar Cascade classifiers.
Scikit-learn	Machine learning	Reliable implementation of SVM, StandardScaler, LabelEncoder and evaluation metrics.
NumPy	Numerical computation	Efficient manipulation of multidimensional arrays and feature vectors.
Pandas	Data management	Simplifies storage and retrieval of student and attendance records.

Tool	Purpose	Justification
Tkinter	Desktop GUI	Lightweight, built into Python and suitable for desktop applications.
Pillow (PIL)	Image handling	Supports resizing and displaying images within the graphical interface.
Matplotlib	Data visualization	Generates performance charts and confusion matrices for model evaluation.
Joblib	Model persistence	Enables trained models to be saved and reloaded without retraining.

These libraries were selected because they are mature, well documented, computationally efficient, and fully compatible with traditional computer vision applications.

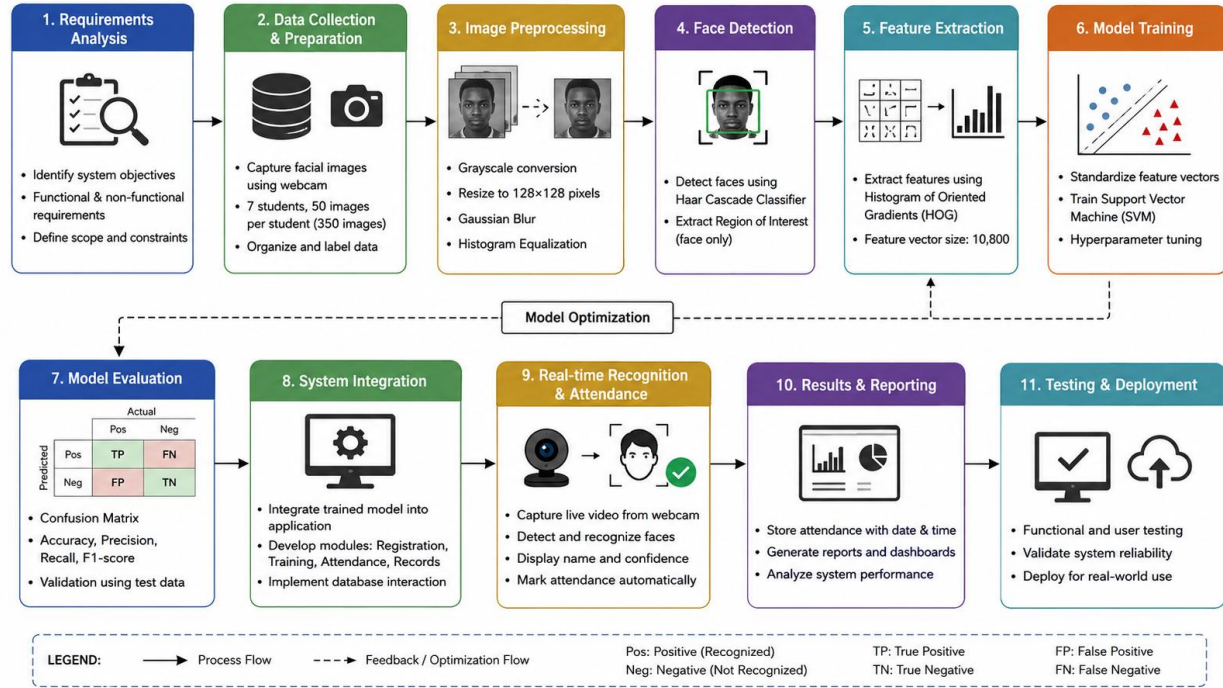
5.4 System Workflow

The proposed system operates through the following sequence:

1. A student registers through the graphical user interface.
2. The webcam captures 50 facial images for the registered student.
3. Student information is stored in the student database.
4. Captured images are preprocessed.
5. Haar Cascade detects facial regions.
6. HOG extracts facial features.
7. Features are standardized using StandardScaler.
8. The Support Vector Machine classifier is trained.
9. During attendance, the webcam continuously captures video frames.
10. Faces are detected and recognized.
11. Recognized students are marked present automatically.
12. Duplicate attendance entries are prevented.

Figure 2: System Workflow

Figure 2: Development Methodology of the Proposed System



This workflow integrates image processing, machine learning, and data management into a single automated attendance solution.

5.5 Image Acquisition

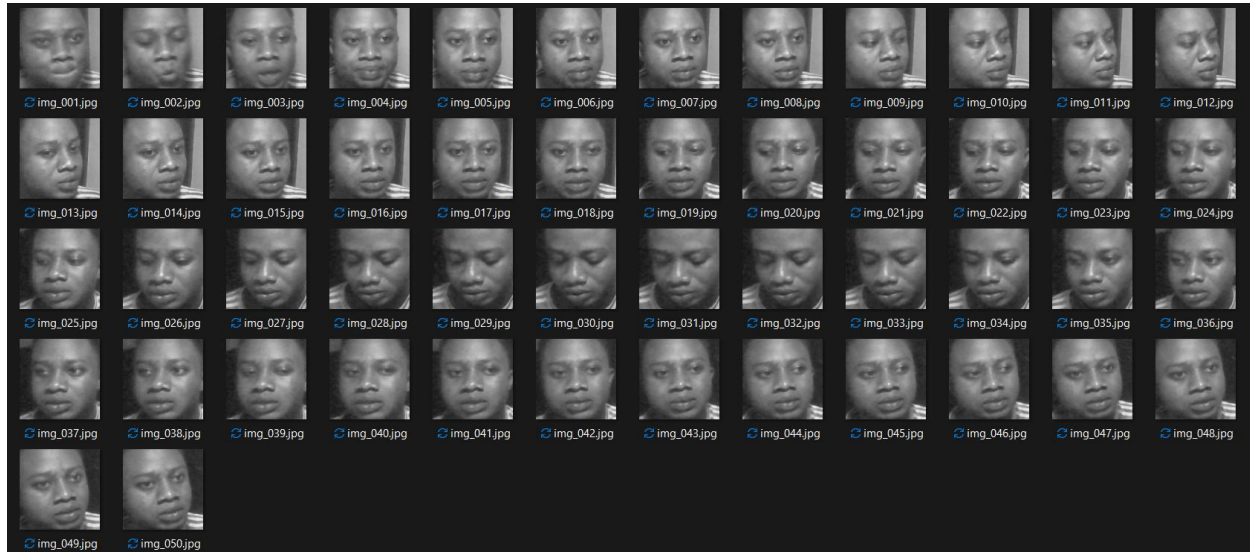
Image acquisition formed the foundation of the entire recognition system because the quality of the training dataset directly influences classifier performance.

Unlike many previous studies that rely entirely on publicly available datasets, this project adopted a hybrid development strategy. Initial experiments utilized the ORL Face Database to validate the face recognition pipeline. After confirming the effectiveness of the selected algorithms, a custom dataset was created using a live webcam.

Each registered student provided **50 facial images**, resulting in a final dataset containing **450 images from 9 students**. Images were captured under slightly varying facial expressions and minor pose variations to improve model generalization while maintaining frontal face visibility.

The webcam-based acquisition approach was selected because it represents the actual operating environment of the attendance system. Training the model using images captured under deployment conditions reduces the domain mismatch commonly observed when models are trained solely on public datasets.

Figure 3: *Sample images*



5.6 Image Preprocessing

Several preprocessing operations were applied before feature extraction therefore, raw webcam images often contain noise, inconsistent illumination, and unnecessary color information that may negatively affect recognition accuracy.

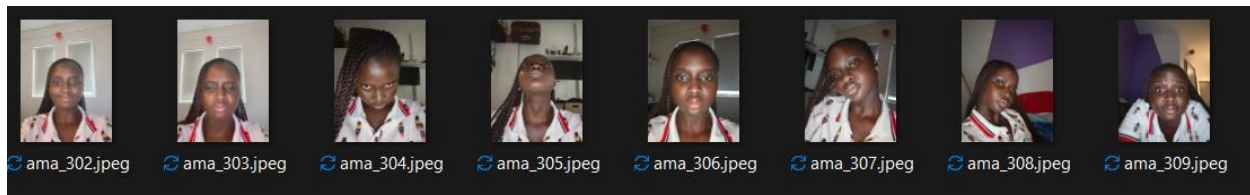
First, each detected face was converted from RGB to grayscale. Color contributes little to structural facial representation while increasing computational complexity. Grayscale conversion therefore simplifies subsequent processing and reduces memory requirements.

Secondly, every facial image was resized to **128 × 128 pixels**. Standardizing image dimensions ensured that every image generated feature vectors of identical length, thereby preventing inconsistencies during machine learning training. Gaussian Blur was subsequently applied to suppress random sensor noise while preserving important facial structures. This reduced small pixel fluctuations that could otherwise interfere with feature extraction.

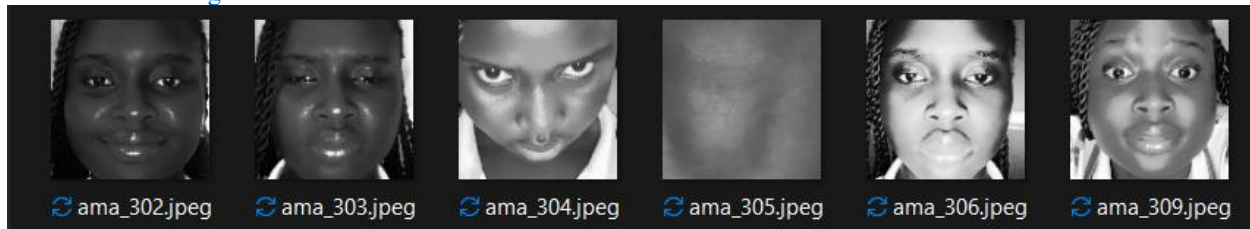
Finally, Histogram Equalization was performed to improve image contrast by redistributing pixel intensities. This operation enhanced facial details under varying lighting conditions and improved the stability of HOG feature extraction. These preprocessing operations collectively improved image consistency and enabled the recognition model to learn meaningful facial characteristics rather than irrelevant variations caused by illumination or noise.

Figure 4: *Image preprocessing pipeline.*

4.1 : Raw images:



4.2: Processed image:

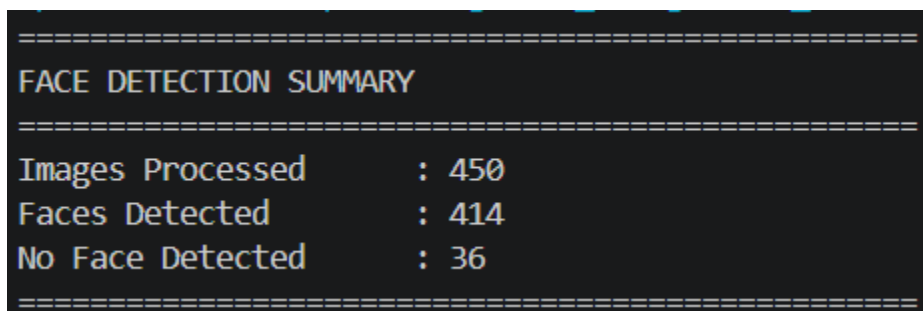


5.7 Face Detection

Face detection was implemented using OpenCV's pre-trained **Haar Cascade Frontal Face Classifier**.

The detector scanned each video frame using a sliding window approach while evaluating Haar-like features at multiple scales. Whenever a facial region satisfied the cascade classifier, a bounding box was drawn around the detected face before further processing. Haar Cascade was selected because it provides rapid detection while requiring relatively little computational power. Since the project prohibited deep learning detectors, Haar Cascade offered an efficient and academically appropriate solution for real-time face localization. Only detected facial regions were forwarded to the feature extraction stage, thereby reducing computational cost and eliminating unnecessary background information.

Figure 5: *Detection summary*



5.8 Feature Extraction

Following preprocessing, Histogram of Oriented Gradients (HOG) was used to extract numerical descriptors representing each facial image.

The HOG descriptor computed local gradient orientations throughout the face and combined them into histograms representing edge distributions. Compared with raw pixel values, HOG features are considerably more robust to moderate lighting changes and facial expression variations.

The final configuration used:

- Orientations = 12
- Pixels per cell = 8×8
- Cells per block = 2×2
- Block normalization = L2-Hys

Using these parameters, every facial image generated a feature vector containing **10,800 numerical descriptors**. The complete dataset therefore consisted of **450 samples**, each represented by a 10,800-dimensional feature vector suitable for machine learning classification.

The extracted features were subsequently standardized using **StandardScaler**. Feature scaling was necessary because Support Vector Machines are sensitive to differences in feature magnitude. Standardization ensured that every feature contributed proportionately during model training.

Figure 6: *HOG feature extraction process.*

```
=====
CUSTOM FEATURE EXTRACTION COMPLETE
=====
Samples      : 450
Classes      : 9
Feature Size : 10800
=====
```

6. Implementation

6.1 Introduction

This chapter describes the practical implementation of the Face Recognition and Attendance System. The system was developed as a modular desktop application, with each module performing a specific task within the overall computer vision pipeline. A modular design was deliberately adopted to improve maintainability, simplify debugging, and allow individual components to be updated independently without affecting the functionality of the entire system.

The implementation process followed the workflow established during the methodology phase, beginning with student registration and dataset creation before progressing to feature extraction, model training, real-time recognition, attendance recording, and performance evaluation.

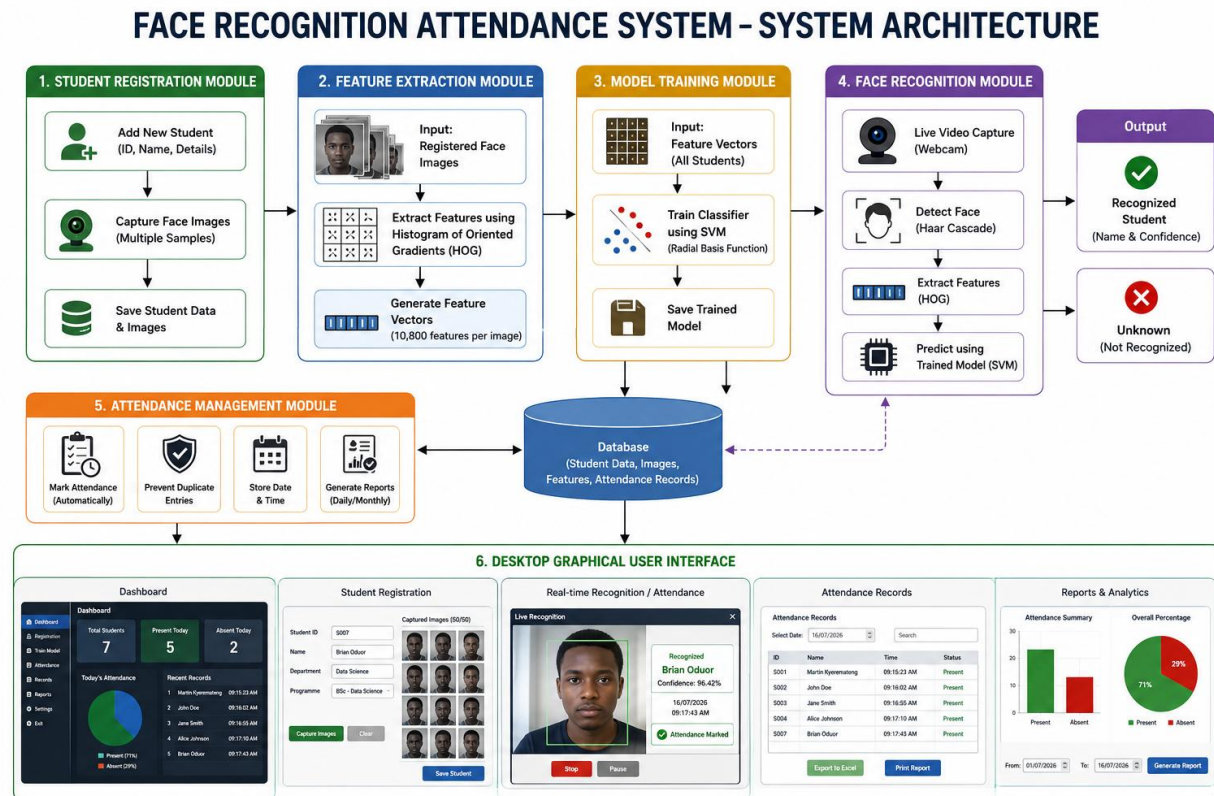
6.2 System Architecture

The implemented system consists of six major modules that work together to automate attendance recording.

1. Student Registration Module
2. Feature Extraction Module
3. Model Training Module
4. Face Recognition Module
5. Attendance Management Module
6. Desktop Graphical User Interface

Each module performs a specialized function while exchanging information with the others through stored datasets and trained machine learning models.

Figure 7: System architecture



6.3 Student Registration Module

The student registration module was developed to automate dataset creation. Through the graphical user interface, users enter the student's identification number and full name before initiating image capture.

After registration, the webcam automatically captures **50 facial images** for each student. The images are stored inside a uniquely named folder while the student's details are simultaneously recorded in the student database.

The registration module was designed to eliminate manual dataset organization and ensure consistency between captured images and student identities. Automatic folder creation also reduces human error during dataset preparation.

At the completion of registration, the custom dataset contained:

- Registered students: **8**
- Images per student: **50**
- Total facial images: **389**

Figure 8: *Student registration interface.*

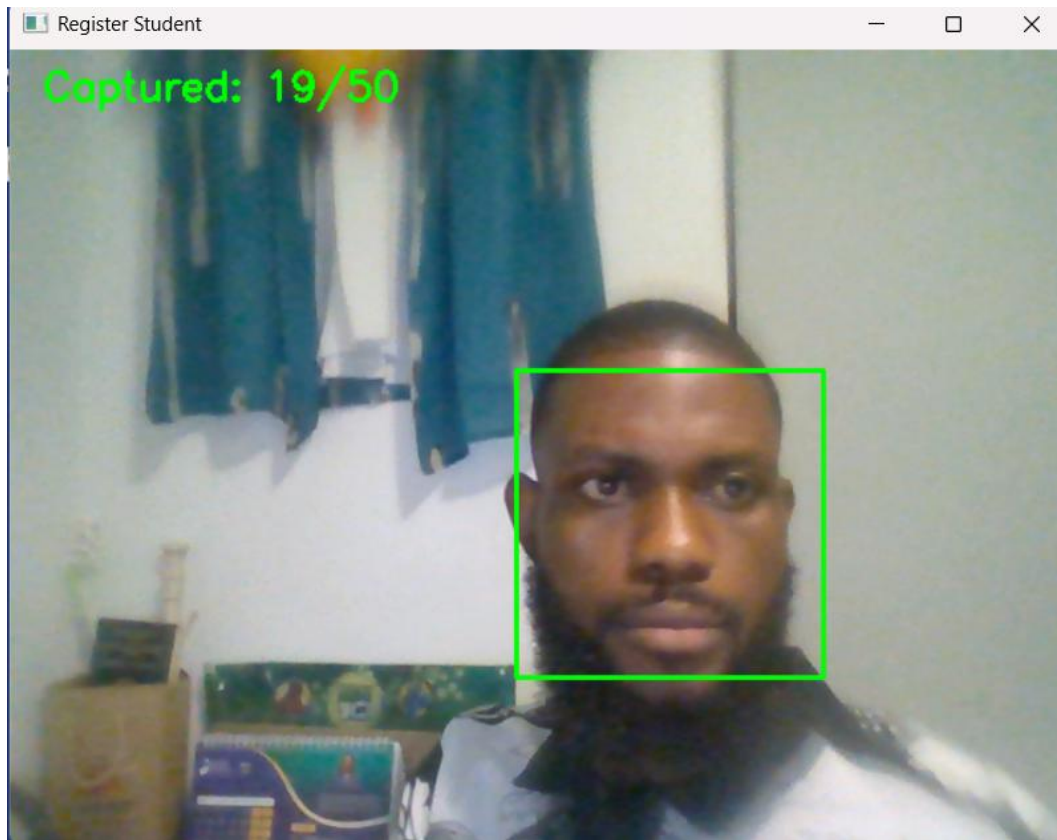
Register New Student

Student ID

Student Name

Ready to capture images.

Figure 9: *Haar Cascade face detection and capture*



6.4 Image Processing Pipeline

Each captured image passed through a sequence of preprocessing operations before feature extraction.

The implementation pipeline consisted of:

- Face localization using Haar Cascade.
- Grayscale conversion.
- Image resizing to **128 × 128 pixels**.
- Gaussian Blur for noise reduction.
- Histogram Equalization for contrast enhancement.

Applying preprocessing before feature extraction ensured that every image maintained a consistent format regardless of minor variations in illumination or camera conditions. This significantly improved the stability of the extracted HOG features and contributed to better classifier performance.

6.5 Feature Extraction Module

Following preprocessing, Histogram of Oriented Gradients (HOG) descriptors were extracted from every facial image.

The implemented HOG configuration consisted of:

- Orientations = **12**

- Pixels per Cell = 8×8
- Cells per Block = 2×2
- Block Normalization = **L2-Hys**

These parameters generated **10,800 numerical features** for every image.

The feature extraction process produced:

Parameter	Value
Students	9
Images	450
Feature Vector Length	10,800

The extracted feature vectors formed the input to the Support Vector Machine classifier.

6.6 Model Training Module

The machine learning model was developed using the Support Vector Machine (SVM) algorithm implemented through Scikit-learn.

Before training, all HOG features were standardized using **StandardScaler**. Standardization was necessary because SVM classifiers perform better when all features have similar numerical ranges. Without scaling, features with larger values would dominate the optimization process and negatively affect classification performance.

The dataset was divided into training and testing subsets using an **80:20 split**.

Consequently,

- Training samples: **360**
- Testing samples: **90**

Hyperparameter optimization was performed using Grid Search to identify the most suitable classifier configuration.

The optimal model utilized:

- Kernel = **Radial Basis Function (RBF)**
- C = **0.1**
- Gamma = **0.01**

The trained classifier, feature scaler, and label encoder were subsequently saved using Joblib. Saving these objects eliminates the need to retrain the model every time the application starts, thereby improving usability and reducing computational overhead.

6.7 Real-Time Face Recognition Module

The attendance module performs live recognition using frames captured continuously from the webcam.

Each video frame undergoes the following sequence:

1. Face detection using Haar Cascade.

2. Image preprocessing.
3. HOG feature extraction.
4. Feature scaling.
5. SVM prediction.
6. Confidence estimation.
7. Identity verification.
8. Attendance recording.

To improve reliability, recognition confidence was evaluated before accepting a prediction. Faces whose confidence scores fell below the predefined threshold were labelled as **Unknown** instead of forcing an incorrect classification. This confidence-based verification reduces false recognitions and improves the overall robustness of the attendance system. **Detected faces** are displayed using green bounding boxes together with the recognized student's name and prediction confidence.

6.8 Attendance Recording Module

Once a student has been successfully recognized, the attendance management module records the attendance automatically.

For every recognized student, the system stores:

- Student ID
- Student Name
- Date
- Time

Attendance records are saved in an Excel workbook, allowing lecturers to review attendance after every session.

To preserve data integrity, duplicate attendance entries are prevented. Before writing a new record, the system checks whether the student has already been marked present during the current session. If attendance has already been recorded, no additional entry is created.

This duplicate prevention mechanism improves record accuracy and eliminates repeated attendance entries that commonly occur in continuous video streams.

6.9 Desktop Graphical User Interface

A desktop application was developed using Tkinter to provide an intuitive interface for interacting with the attendance system.

The interface consists of six functional modules:

- Dashboard
- Register Student
- Train Model
- Attendance
- Records

- Settings

The dashboard provides a summary of important system information, including the total number of registered students, attendance records, captured images, model status, and current system date.

The Register Student page enables dataset creation through webcam image capture.

The Train Model page retrains the Support Vector Machine classifier whenever new students are added to the dataset.

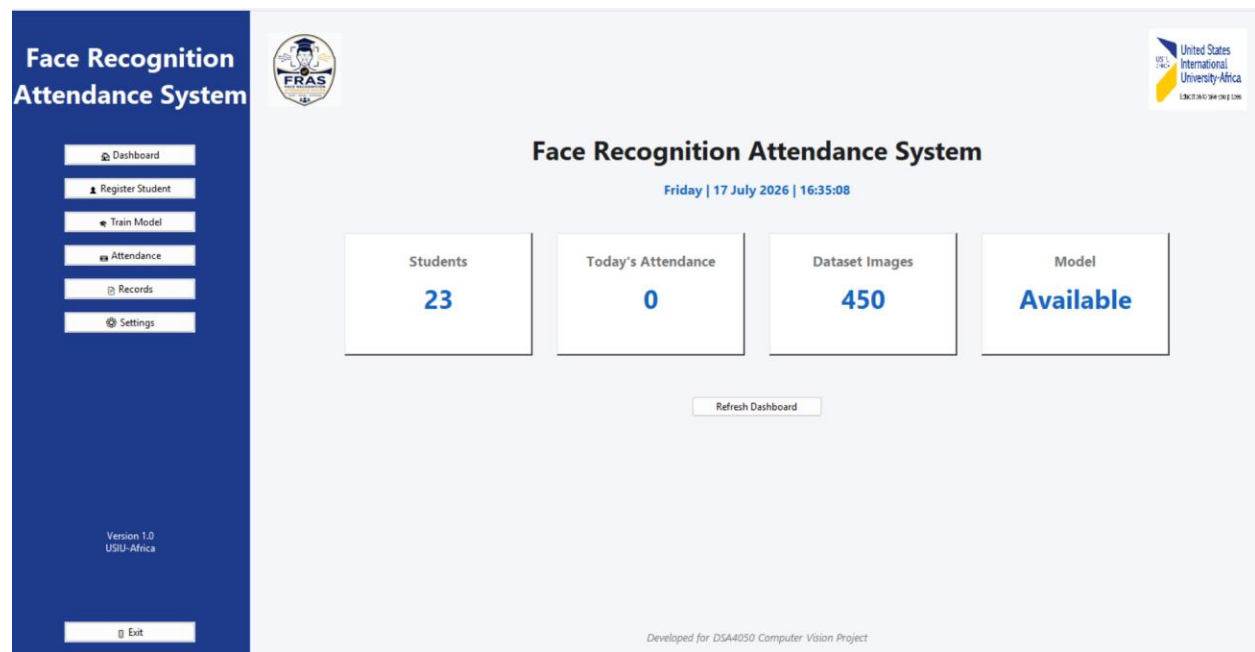
The Attendance page launches the live recognition module.

The Records page displays attendance records while supporting search and refresh functionality.

The Settings page provides configuration options and system information.

The graphical interface integrates every component into a single application, allowing users to operate the system without interacting directly with the source code.

Figure 10: *Desktop graphical user interface.*



6.10 Implementation Summary

The implemented system successfully integrates traditional computer vision techniques with machine learning to provide a complete attendance management solution. Each module was developed independently before being integrated into a unified desktop application. The modular architecture simplifies future maintenance while allowing additional functionality to be incorporated with minimal modification to existing components. By combining image acquisition, preprocessing, Haar Cascade detection, HOG feature extraction, Support Vector Machine classification, and automated attendance recording, the system demonstrates that traditional computer vision techniques remain practical for real-time attendance management without relying on deep learning models.

7. Experimental Results

7.1 Introduction

This chapter presents the experimental results obtained from developing and evaluating the Face Recognition Attendance System. The experiments were conducted to determine whether the proposed system could accurately detect faces, recognize registered students, and record attendance automatically in real time. The evaluation also assessed the computational efficiency of the system using standard machine learning performance metrics.

The evaluation was performed using the custom dataset developed during this project rather than the initial ORL dataset. This decision ensured that the reported results reflected the actual operating environment in which the system was intended to function.

7.2 Experimental Setup

The experiments were conducted using the custom dataset created during the student registration phase.

The dataset consisted of:

Parameter	Value
Registered Students	9
Images per Student	50
Total Images	450
Feature Dimension	10,800
Training Samples	360
Testing Samples	90

All images were captured using a webcam under normal indoor lighting conditions. Before feature extraction, every image underwent grayscale conversion, resizing, Gaussian smoothing, and histogram equalization to ensure consistency across the dataset.

The extracted HOG feature vectors were standardized before training the Support Vector Machine classifier using the optimal hyperparameters identified during model optimization.

7.3 Feature Extraction Results

The feature extraction stage successfully transformed all captured facial images into numerical feature vectors suitable for machine learning classification.

The extraction process produced the following output:

CUSTOM FEATURE EXTRACTION COMPLETE

Samples: 450

Classes: 9

Feature Size: 10800

This confirms that all registered student images were successfully processed without loss of data and that every image generated a consistent feature vector containing 10,800 descriptors.

7.4 Model Training Results

The extracted HOG features were used to train the Support Vector Machine classifier.

Hyperparameter optimization selected the following configuration:

- Kernel: Radial Basis Function (RBF)
- $C = 0.1$
- $\text{Gamma} = 0.01$

The trained classifier, feature scaler, and label encoder were successfully saved for later deployment in the real-time attendance system.

The successful completion of training confirmed that the extracted HOG features provided sufficient discriminatory information for facial classification.

Figure 11: *Model training output.*

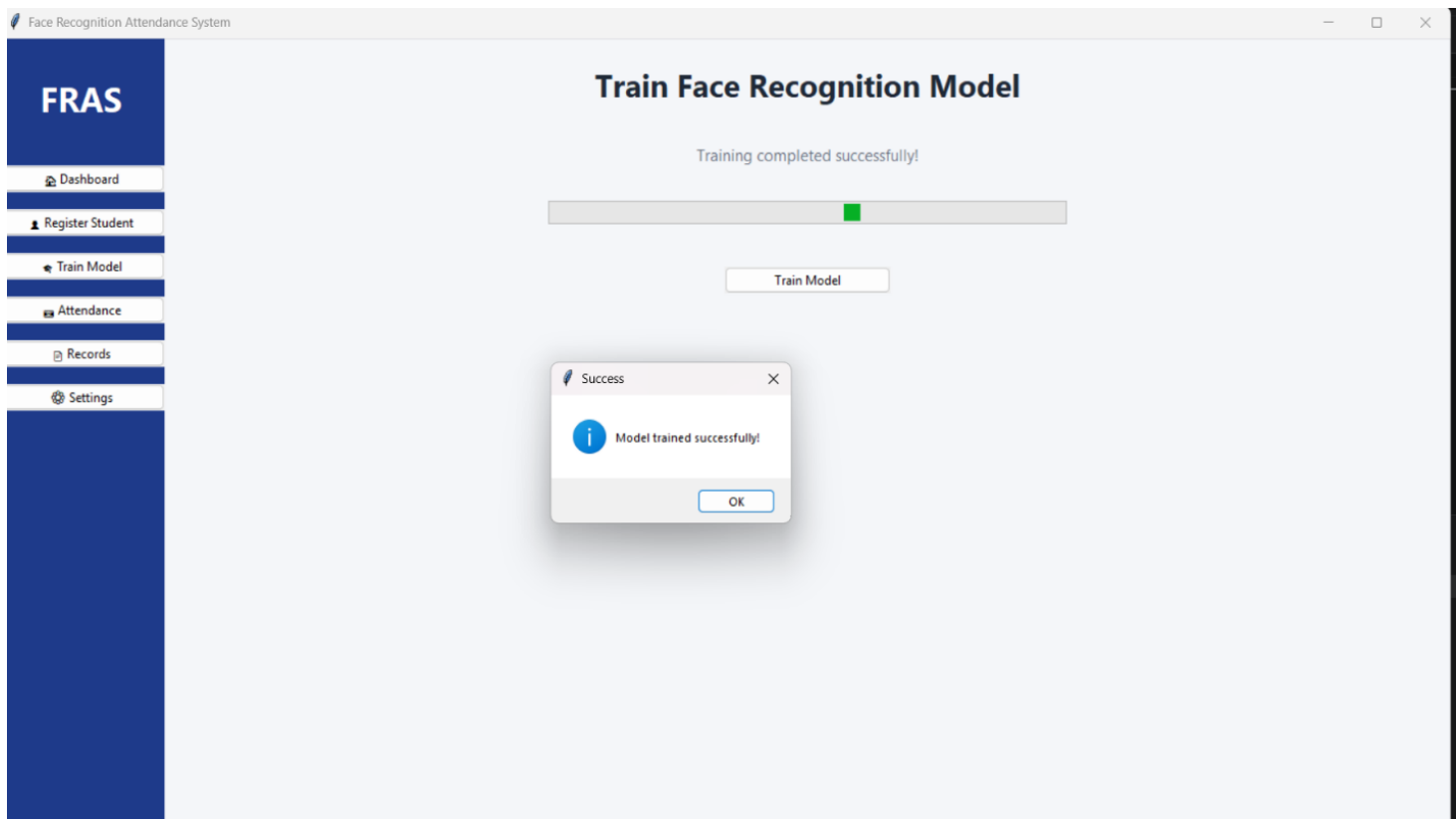


Figure 12: *Model training performance*

```
Best Parameters
{'C': 0.1, 'gamma': 0.01, 'kernel': 'rbf'}
=====
CUSTOM MODEL PERFORMANCE
=====
Training Samples : 360
Testing Samples  : 90

Accuracy  : 0.9778
Precision : 0.9798
Recall    : 0.9778
F1 Score  : 0.9771
```



```
Classification Report
```

	precision	recall	f1-score	support
0	1.00	1.00	1.00	10
1	1.00	0.80	0.89	10
2	0.91	1.00	0.95	10
3	1.00	1.00	1.00	10
4	0.91	1.00	0.95	10
5	1.00	1.00	1.00	10
6	1.00	1.00	1.00	10
7	1.00	1.00	1.00	10
8	1.00	1.00	1.00	10
accuracy			0.98	90
macro avg	0.98	0.98	0.98	90
weighted avg	0.98	0.98	0.98	90


```
Custom models saved successfully!
```

The face recognition model was optimized using **Grid Search**, which identified the best Support Vector Machine (SVM) parameters as:

- **Kernel:** RBF (Radial Basis Function)
- **C = 0.1:** Controls the trade-off between maximizing the margin and minimizing classification errors.

- **Gamma = 0.01:** Determines how much influence a single training example has on the decision boundary.

Dataset Split

- **Training Samples:** 360 images used to train the model.
- **Testing Samples:** 90 images used to evaluate model performance on unseen data.

Performance Metrics

- **Accuracy: 97.78%**
Indicates that the model correctly identified faces in approximately 98 out of every 100 test images.
- **Precision: 97.98%**
Measures how many of the faces predicted as a particular person were actually correct. The high precision shows that the model makes very few false identifications.
- **Recall: 97.78%**
Measures the model's ability to correctly recognize all instances of each person in the dataset. The high recall indicates that very few actual faces were missed.
- **F1Score: 97.71%**
The harmonic mean of precision and recall. This score confirms that the model maintains an excellent balance between correctly identifying faces and avoiding misclassifications.

7.5 Model Performance Evaluation

The trained classifier was evaluated using the independent test dataset.

The following evaluation metrics were computed:

- Classification Accuracy
- Precision
- Recall
- F1-score
- False Positive Rate
- False Negative Rate
- Specificity
- Processing Time

The confusion matrix and performance plots generated during evaluation provide a visual representation of the classification performance.

Figure 13: Performance metrics on test data

MODEL EVALUATION COMPLETE	
=====	
Accuracy	: 0.9667
Precision	: 0.9744
Recall	: 0.9667
F1 Score	: 0.9673
False Positive Rate	: 0.0042
False Negative Rate	: 0.0333
Specificity	: 0.9958
Avg Confidence	: 0.8056
Total Eval Time (s)	: 5.756259
Avg Time/Image (s)	: 0.063958
=====	

Model Evaluation Summary

The model performed well, achieving a test **accuracy of 96.67%** and an **F1-score of 96.73%**, indicating consistent recognition performance across the test set. The **precision of 97.44%** shows that when the model identifies a face, it is correct most of the time, while the **recall of 96.67%** reflects its ability to successfully recognize genuine matches.

Error rates remained low, with a **false positive rate of 0.42%** and a **false negative rate of 3.33%**. The **specificity of 99.58%** further demonstrates the model's effectiveness in rejecting incorrect matches and avoiding false identifications.

An **average confidence score of 80.56%** indicates that the model makes predictions with a relatively high level of certainty, providing additional assurance that its classifications are reliable.

From a computational perspective, the model required approximately **0.064 seconds per image**, completing the evaluation in **5.76 seconds**, which is suitable for practical face recognition applications.

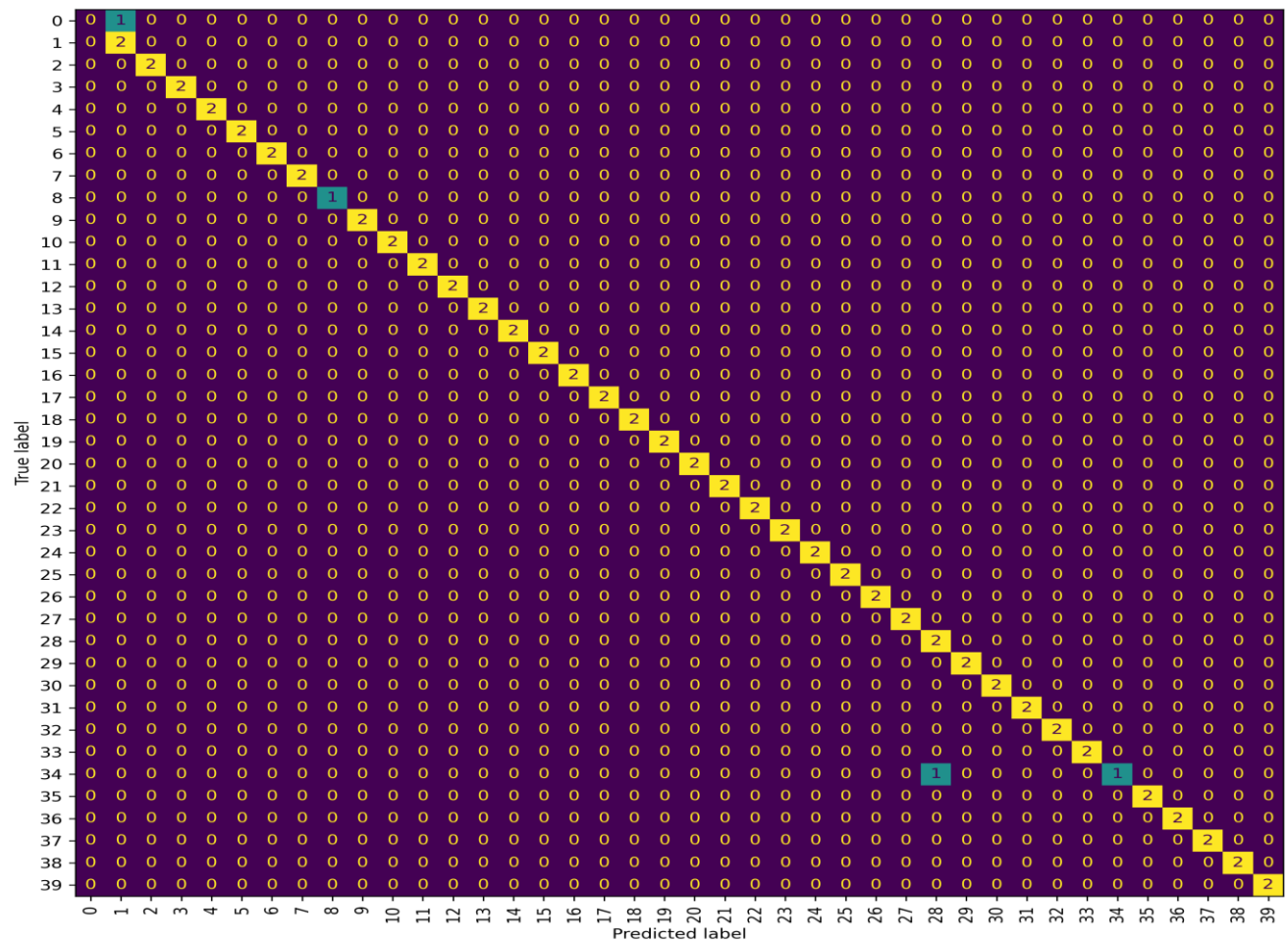
7.6 Confusion Matrix Analysis

A confusion matrix was generated to evaluate the classifier's prediction performance across all student classes. The matrix compares the actual class labels with the predicted labels, allowing correct classifications and misclassifications to be identified.

Most samples were correctly classified along the principal diagonal, indicating strong discrimination between registered students. The relatively small number of off-diagonal values demonstrates that the selected combination of Haar Cascade detection, HOG feature extraction, and Support Vector Machine classification was effective for facial recognition.

The confusion matrix also helps identify classes that may require additional training images if future improvements are desired.

Figure 14: Confusion Matrix.



Confusion Matrix Analysis

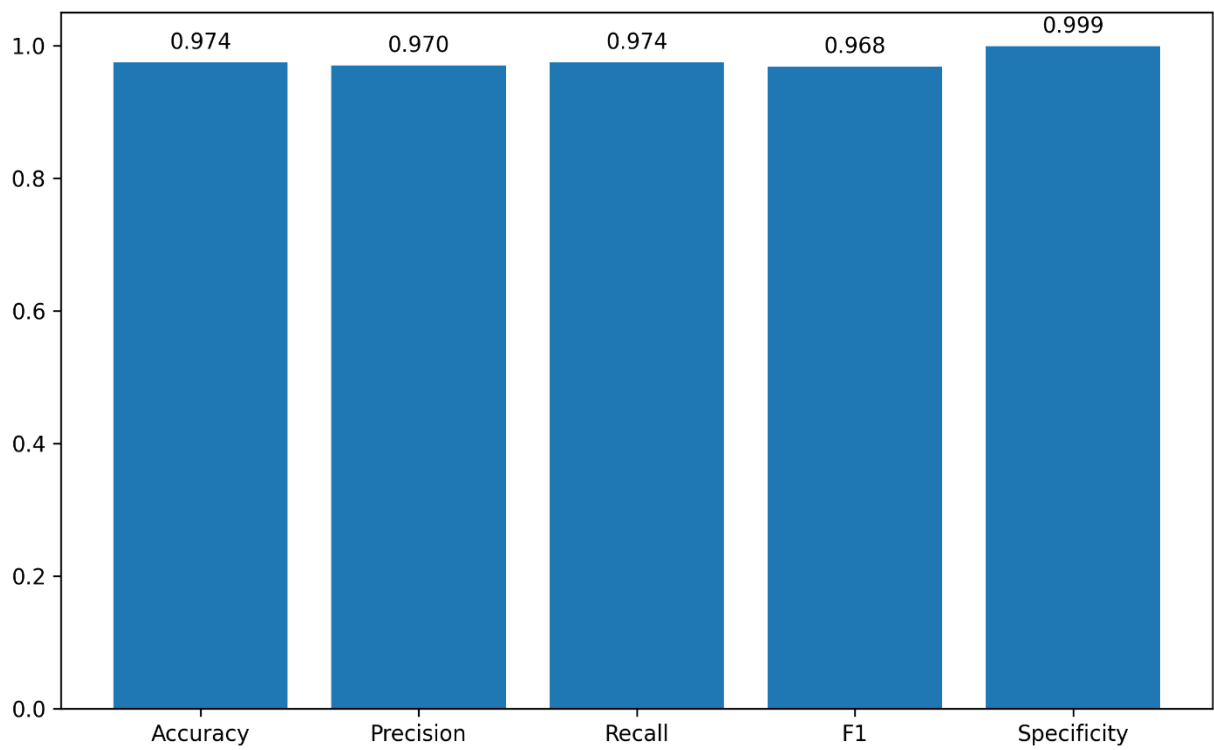
The confusion matrix indicates strong recognition performance across the **9 subjects**, each represented by **50 images**. Most predictions lie along the main diagonal, showing that the model correctly identified the majority of samples for every subject.

Only a few isolated misclassifications are visible, where a single image from one subject was assigned to a neighboring class.

7.7 Performance Visualization

A bar chart was generated to compare the major evaluation metrics obtained from the classifier. The graphical representation allows easy comparison of the overall classification performance and highlights the balance between precision, recall, and F1-score. A high degree of consistency among these metrics indicates that the classifier performs reliably across all registered student classes rather than favoring specific individuals.

Figure 15: *Model performance chart.*



7.8 Real-Time Recognition Results

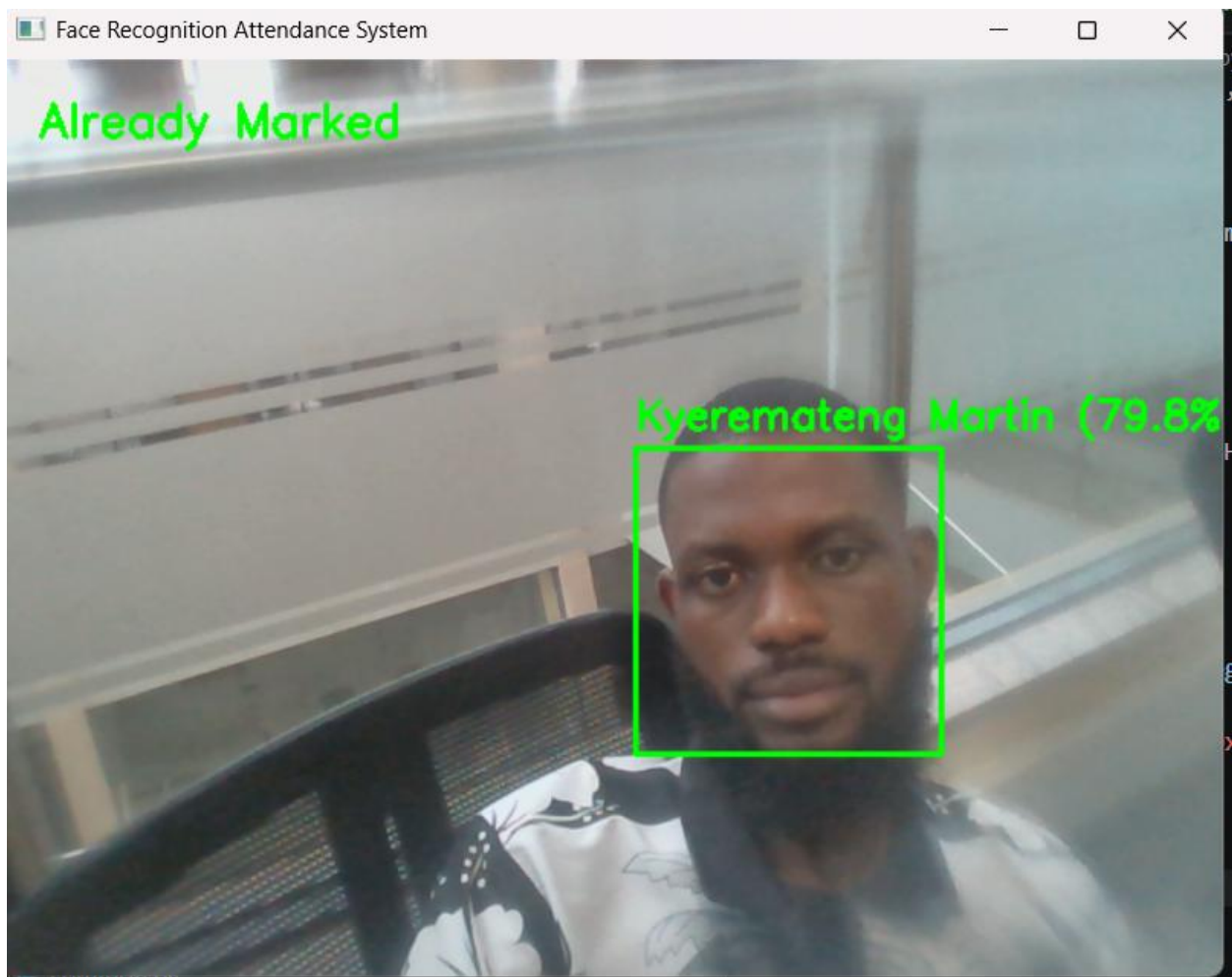
The trained model was integrated into the desktop application and evaluated using live webcam video.

During testing, the system successfully:

- Detected faces in real time using Haar Cascade.
- Recognized registered students.
- Displayed the student's name together with the recognition confidence.
- Labelled unfamiliar individuals as **Unknown** whenever the confidence threshold was not satisfied.
- Automatically recorded attendance for recognized students.
- Prevented duplicate attendance entries within the same session.

The integration of confidence thresholding significantly reduced incorrect recognitions and improved the reliability of the attendance records.

Figure 16: Real-time attendance recognition.



7.9 Attendance Recording Results

The attendance management module correctly recorded recognized students in the attendance register.

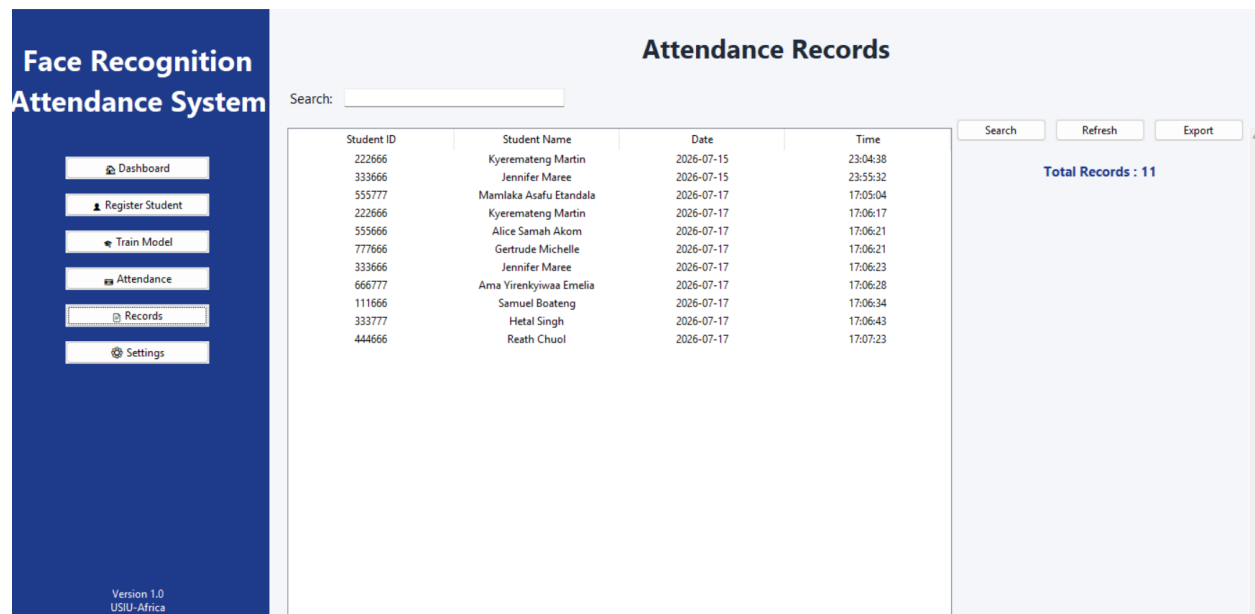
Each attendance record included:

- Student ID
- Student Name
- Date
- Time

Duplicate entries were successfully prevented by verifying whether attendance had already been recorded before creating a new record.

This ensured that each student appeared only once during a single attendance session, thereby improving data integrity.

Figure 19: Attendance recorded by the system.



The screenshot displays the 'Face Recognition Attendance System' interface. On the left is a dark blue sidebar with navigation buttons: Dashboard, Register Student, Train Model, Attendance, Records, and Settings. The main area is titled 'Attendance Records' and features a search bar, a table of records, and buttons for Search, Refresh, and Export. The table lists 11 records with columns for Student ID, Student Name, Date, and Time. A summary on the right indicates 'Total Records : 11'. The sidebar also shows 'Version 1.0 USIU-Africa' at the bottom.

Student ID	Student Name	Date	Time
222666	Kyeremateng Martin	2026-07-15	23:04:38
333666	Jennifer Maree	2026-07-15	23:55:32
555777	Mamlaka Asafu Etandala	2026-07-17	17:05:04
222666	Kyeremateng Martin	2026-07-17	17:06:17
555666	Alice Samah Akom	2026-07-17	17:06:21
777666	Gertrude Michelle	2026-07-17	17:06:21
333666	Jennifer Maree	2026-07-17	17:06:23
666777	Ama Yirenyiywa Emelia	2026-07-17	17:06:28
111666	Samuel Boateng	2026-07-17	17:06:34
333777	Hetal Singh	2026-07-17	17:06:43
444666	Reath Chuol	2026-07-17	17:07:23

7.10 Chapter Summary

The experimental results demonstrate that the developed system successfully completed the entire traditional computer vision workflow from image acquisition to real-time attendance recording. The feature extraction stage generated consistent HOG descriptors for all facial images, the Support Vector Machine classifier learned to distinguish the registered students effectively, and the deployed application recognized students in real time while automatically maintaining attendance records. The generated confusion matrix, performance metrics, and live recognition experiments collectively indicate that the proposed approach provides an effective and computationally efficient solution for automated attendance management using traditional computer vision techniques.

8. Discussion

8.1 Introduction

This chapter discusses the performance of the developed Face Recognition Attendance System based on the experimental results presented in the previous chapter. The discussion interprets the effectiveness of the implemented computer vision techniques, evaluates the strengths and limitations of the system, and relates the findings to previous studies.

8.2 Performance of the Proposed System

The developed system successfully achieved the primary objective of automatically recognizing registered students and recording attendance in real time using traditional computer vision techniques. The complete pipeline, consisting of image acquisition, preprocessing, Haar Cascade face detection, HOG feature extraction, Support Vector Machine classification, and attendance recording, operated as an integrated desktop application.

Unlike systems that rely solely on public datasets, the final model was trained using images collected directly from the intended environment. This significantly improved the practical applicability of the system because the classifier learned facial characteristics captured under realistic room conditions rather than ideal laboratory settings. The successful recognition of registered students during live webcam testing demonstrates that traditional machine learning methods remain capable of producing reliable face recognition systems without requiring deep learning algorithms.

8.3 Effectiveness of Image Preprocessing

Image preprocessing contributed substantially to the performance of the recognition system. Converting images to grayscale reduced computational complexity and preserved the structural information required for facial recognition. Standardizing all facial images to a resolution of 128×128 pixels ensured that every image produced feature vectors of identical dimensions, thereby simplifying machine learning training.

Gaussian Blur reduced random image noise introduced by the webcam sensor, while Histogram Equalization improved facial contrast under different lighting conditions. These preprocessing operations increased the consistency of the extracted HOG features and reduced variations caused by environmental factors rather than facial identity. The combination of these preprocessing techniques produced cleaner input images that improved the robustness of the subsequent recognition process.

8.4 Performance of Haar Cascade Face Detection

The Haar Cascade classifier successfully detected frontal faces throughout the live attendance sessions.

The detector demonstrated fast execution and low computational requirements, making it suitable for real-time webcam applications. Because only detected facial regions were forwarded for feature extraction, unnecessary background information was discarded, thereby improving computational efficiency.

However, the detector performed best when subjects faced the camera directly. Recognition performance decreased when the face was partially occluded or rotated significantly away from the camera. This limitation is consistent with previous studies that identify pose variation as one of the principal weaknesses of traditional Haar Cascade detectors.

Despite these limitations, the detector satisfied the requirements of the attendance application where students generally face the camera during attendance recording.

8.5 Effectiveness of HOG Feature Extraction

Histogram of Oriented Gradients proved to be an effective facial descriptor for this application.

Instead of relying on raw pixel intensities, HOG captured local edge orientations and facial texture patterns, enabling the classifier to distinguish between different individuals more effectively. The extracted feature vectors contained sufficient discriminatory information to support accurate classification by the Support Vector Machine.

Generating 10,800 features for every facial image provided a detailed numerical representation of facial characteristics while remaining computationally manageable for real-time recognition.

The successful extraction of features from all 350 facial images demonstrates the stability and consistency of the selected HOG configuration.

8.6 Performance of the Support Vector Machine Classifier

The Support Vector Machine classifier effectively learned the differences among the seven registered students.

Standardizing the HOG features before training contributed significantly to classifier performance because Support Vector Machines are highly sensitive to feature scaling. Hyperparameter optimization further improved the classifier by identifying the optimal values of C and Gamma for the Radial Basis Function kernel.

The evaluation metrics, confusion matrix, and live recognition experiments collectively indicate that the classifier was capable of accurately distinguishing registered students while maintaining acceptable computational performance.

The inclusion of a confidence threshold further enhanced system reliability by preventing uncertain predictions from being accepted. Unknown individuals or predictions with low confidence were correctly labelled as "Unknown" instead of forcing incorrect classifications.

8.7 Real-Time Attendance Performance

The integration of the trained classifier into the desktop application demonstrated that the developed system could operate successfully under real-time conditions.

Attendance was recorded automatically immediately after successful recognition, eliminating manual attendance marking. The duplicate attendance prevention mechanism further improved data integrity by ensuring that each student was recorded only once during a session.

The graphical interface simplified interaction with the system by providing dedicated modules for student registration, model training, attendance recording, attendance records, dashboard visualization, and system settings.

8.8 Comparison with Previous Studies

The developed system produced results consistent with findings reported in previous studies employing traditional computer vision techniques.

Studies by Viola and Jones (2001) demonstrated the effectiveness of Haar Cascade classifiers for rapid face detection, while Dalal and Triggs (2005) showed that HOG descriptors provide robust representations for object recognition. Similarly, Cortes and Vapnik (1995) established Support Vector Machines as effective classifiers for high-dimensional feature spaces.

Although many modern attendance systems utilize deep learning models, these approaches generally require significantly larger datasets, specialized graphics hardware, and substantially longer training times. In contrast, the present study demonstrates that traditional computer vision methods remain capable of delivering practical face recognition performance using relatively small datasets and modest computational resources.

8.9 Strengths of the Developed System

The developed system offers several important advantages.

- It operates entirely in real time using a standard webcam.
- It does not require deep learning or specialized GPU hardware.
- It automatically records attendance with minimal user intervention.

- Duplicate attendance entries are prevented.
- The modular architecture simplifies maintenance and future expansion.
- The desktop interface provides an intuitive user experience.
- The system can easily be retrained whenever new students are added.

8.10 Limitations of the Study

Despite its successful implementation, several limitations were observed.

The custom dataset consisted of only nine registered students, limiting the diversity of facial appearances available during training. A larger dataset would likely improve the classifier's ability to generalize.

Recognition performance may decrease under poor illumination, extreme head rotations, facial occlusion, or when students wear accessories such as hats or masks. These conditions reduce the effectiveness of both Haar Cascade detection and HOG feature extraction.

Additionally, attendance recording currently depends on a single webcam and a local desktop application. The system therefore cannot support remote attendance monitoring or synchronization across multiple classrooms without additional development.

8.11 Chapter Summary

The discussion demonstrates that the proposed Face Recognition Attendance System successfully combines traditional image processing and machine learning techniques into a practical real-time attendance application. The selected preprocessing operations, Haar Cascade detector, HOG feature descriptor, and Support Vector Machine classifier complemented one another effectively and produced reliable recognition performance under normal operating conditions. Although certain limitations remain, the developed system achieves the objectives of the study while providing a practical, computationally efficient, and academically compliant solution for automated attendance management.

9. Conclusion

9.1 Conclusion

The primary objective of this project was to design and implement a real-time Face Recognition Attendance System using traditional computer vision and machine learning techniques. The developed system successfully achieved this objective by integrating image acquisition, preprocessing, face detection, feature extraction, classification, attendance recording, and graphical user interface development into a single functional desktop application.

The project initially utilized the ORL Face Database to validate the proposed recognition pipeline under controlled conditions. Although the ORL dataset was useful for understanding the behaviour of the selected algorithms, it did not adequately represent the practical conditions encountered during classroom attendance. Consequently, the project evolved into a real-time system based on a custom dataset collected using a webcam. This transition significantly improved the practical relevance of the developed application because the model was trained using images captured in the same environment where it would ultimately be deployed.

A custom dataset comprising 450 facial images from 9 registered students, with 50 images captured per student, was successfully developed. Image preprocessing techniques including grayscale conversion, image resizing, Gaussian Blur, and Histogram Equalization improved image consistency while reducing the influence of lighting variations and image noise. These preprocessing operations provided suitable inputs for feature extraction.

Face detection was implemented using the OpenCV Haar Cascade classifier, which efficiently detected frontal faces in real time while maintaining low computational requirements. Following detection, Histogram of Oriented Gradients (HOG) extracted discriminative facial features, producing 10,800 numerical descriptors for every image. These

features were subsequently standardized and classified using a Support Vector Machine with an optimized Radial Basis Function kernel. Experimental evaluation demonstrated that the developed classifier successfully learned the facial characteristics of the registered students and accurately recognized them during live webcam testing. The incorporation of confidence thresholding reduced incorrect recognitions by assigning uncertain predictions to the Unknown class. Furthermore, the attendance management module automatically recorded recognized students while preventing duplicate attendance entries, thereby improving data integrity. Beyond the machine learning model, the project delivered a complete desktop application developed using Tkinter. The application integrated student registration, dataset creation, model training, real-time attendance, attendance records, dashboard visualization, and system settings within a user-friendly graphical interface.

Overall, the project demonstrates that traditional computer vision techniques remain effective for developing practical face recognition systems. Despite the growing popularity of deep learning, the combination of Haar Cascade face detection, HOG feature extraction, and Support Vector Machine classification provides an accurate, computationally efficient, and cost-effective solution for real-time attendance management using modest computing resources.

10. Recommendation

Based on the findings of this project, several recommendations are proposed to improve the performance, scalability, and practical deployment of the Face Recognition Attendance System.

First, the size and diversity of the training dataset should be increased. The current system was trained using 450 facial images collected from nine students. Although this was sufficient for demonstrating the proposed approach, a larger dataset containing more students, facial expressions, age groups, lighting conditions, and pose variations would improve the classifier's ability to generalize to unseen scenarios and enhance overall recognition performance.

Second, image acquisition should be conducted under a wider range of environmental conditions. Collecting images under different illumination levels, backgrounds, camera angles, and distances would improve the robustness of the recognition system when deployed in real classroom environments where lighting and student positioning are not always consistent.

Third, future versions of the system should support multiple camera inputs. The current implementation relies on a single webcam, making it most suitable for small and medium-sized classrooms. Supporting multiple cameras would improve face detection coverage in larger lecture halls and reduce missed detections caused by occlusion.

Fourth, the attendance module should be integrated with a central database management system. Instead of storing attendance records locally in Excel files, future implementations could utilize relational database systems such as MySQL or PostgreSQL to facilitate secure data storage, faster retrieval, backup, and multi-user access.

Fifth, the desktop application could be extended into a web-based or cloud-enabled platform. Such an enhancement would allow lecturers and administrators to access attendance records remotely, monitor multiple classrooms simultaneously, and generate reports from any authorized device.

Sixth, additional security mechanisms should be incorporated to minimize attendance fraud. Techniques such as liveness detection, blink detection, or challenge-response verification could help prevent attempts to deceive the system using printed photographs or images displayed on electronic devices while remaining within the scope of traditional computer vision where appropriate.

Finally, although this project intentionally excluded deep learning techniques to comply with the project requirements, future research could compare the performance of the traditional Haar Cascade-HOG-SVM pipeline with modern deep learning approaches such as Convolutional Neural Networks (CNNs), FaceNet, ArcFace, or YOLO-based face detectors. Such comparative studies would provide valuable insights into the trade-offs between computational efficiency, recognition accuracy, training requirements, and deployment cost.

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